GitHub Actions Cheat Sheet

Workflow syntax, runners and actions as of November 2026 · AI-generated summary — verify against the GitHub docs · see the GitHub Actions pages

WORKFLOW — SKELETON

.github/workflows/ci.yml

```
name: ci
on:
  push:
    branches: [main]
  pull_request:
  workflow_dispatch:
permissions:
  contents: read
jobs:
  build:
    runs-on: ubuntu-latest
    timeout-minutes: 15
    steps:
      - uses: actions/checkout@v7
      - run: npm ci && npm test
```

Serialize or cancel overlapping runs

```
concurrency:
  group: publish
  cancel-in-progress: true
```

Note

A step has either uses or run, never both.

TRIGGERS — FILTER

push / pull_request filters

```
on:
  push:
    branches: [main, 'release/**']
    tags: ['v*.*.*']
    paths-ignore: ['docs/**']
```

schedule (UTC, default branch only)

```
  schedule:
    - cron: '0 3 * * *'
    - cron: '30 5 * * 1-5'  # weekdays
```

workflow_dispatch with typed inputs

```
  workflow_dispatch:
    inputs:
      target:
        type: choice
        options: [staging, production]
```

Others

workflow_call makes a workflow reusable; release and workflow_run chain runs; pull_request_target has secrets, so never run PR code in it.

EXPRESSIONS — READ

Contexts

```
${{ github.sha }}  ${{ github.ref_name }}
${{ secrets.NAME }}  ${{ vars.NAME }}
${{ needs.job.outputs.x }}  ${{ matrix.os }}
${{ hashFiles('**/package-lock.json') }}
```

Conditions with status functions

```
if: github.ref == 'refs/heads/main'
if: always()   # run after a failure too
# success() failure() always() cancelled()
```

Passing data from a run step

```
echo "version=1.2.3" >> "$GITHUB_OUTPUT"
echo "NAME=value" >> "$GITHUB_ENV"
echo "### Done" >> "$GITHUB_STEP_SUMMARY"
echo "::add-mask::$TOKEN"
```

JOBS — CHAIN

needs and job outputs

```
jobs:
  prepare:
    runs-on: ubuntu-latest
    outputs:
      v: ${{ steps.v.outputs.version }}
    steps:
      - id: v
        run: |
          echo "version=1.2.3" \
            >> "$GITHUB_OUTPUT"
  use:
    needs: prepare
    runs-on: ubuntu-latest
    steps:
      - run: echo "$V"
        env:
          V: ${{ needs.prepare.outputs.v }}
```

Note

Outputs are always strings; use artifacts for files. Cycles in needs are rejected.

MATRIX — EXPAND

strategy.matrix

```
runs-on: ${{ matrix.os }}
strategy:
  fail-fast: false
  max-parallel: 4
  matrix:
    os: [ubuntu-latest, windows-latest]
    java: [17, 21, 25]
    include:
      - os: ubuntu-24.04-arm
        java: 21
    exclude:
      - os: windows-latest
        java: 17
```

Use the matrix value

```
- uses: actions/setup-java@v6
  with:
    distribution: temurin
    java-version: ${{ matrix.java }}
```

CACHE & ARTIFACTS — REUSE

actions/cache

```
- uses: actions/cache@v6
  with:
    path: ~/.npm
    key: npm-${{ runner.os }}-${{
      hashFiles('package-lock.json') }}
    restore-keys: |
      npm-${{ runner.os }}-
```

Hand files to a later job

```
- uses: actions/upload-artifact@v7
  with:
    name: site
    path: build/site
    retention-days: 7
- uses: actions/download-artifact@v8
  with: { name: site, path: site }
```

Note

setup-node@v7 with cache: npm caches the npm store without a separate cache step.

CREDENTIALS — SCOPE

Least-privilege token and OIDC

```
permissions:
  contents: read
  id-token: write   # request OIDC token
```

Deployment environment

```
deploy:
  environment:
    name: production
    url: https://example.com
  runs-on: ubuntu-latest
  steps:
    - run: ./deploy.sh
```

OIDC cloud login, no stored keys

aws-actions/configure-aws-credentials@v6, google-github-actions/auth@v3, azure/login@v3. Unlisted permission scopes become none.

REUSE — CALL

Reusable workflow (on: workflow_call)

```
jobs:
  build:
    uses: ./.github/workflows/build.yml
    with:
      node-version: '24'
    secrets: inherit
```

.github/actions/setup/action.yml

```
runs:
  using: composite
  steps:
    - uses: actions/setup-node@v7
      with:
        node-version: '24'
    - run: npm ci
      shell: bash
```

Note

Remote: uses: org/repo/.github/workflows/x.yml@v1. Every composite run step needs shell; a local action needs checkout first.

RUNS & SECURITY — HARDEN

Runner labels

```
ubuntu-latest  ubuntu-24.04-arm
windows-latest windows-11-arm
macos-latest   macos-26-intel
[self-hosted, linux, x64]
```

gh CLI

```
gh workflow run publish.yml -f debug=true
gh run list --workflow publish.yml
gh run watch
gh run view --log-failed
gh run rerun --failed
```

Pin SHAs; pass untrusted text via env

```
- uses: actions/checkout@<sha>  # v7.0.1
- run: echo "Title: $PR_TITLE"
  env:
    PR_TITLE: ${{
      github.event.pull_request.title }}
```